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Roll Number: 62		LAB Assignment No: 1
Title of LAB Assignment: To understand DevOps: Principles, Practices, and DevOps Engineer Role and Responsibilities.		
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PRACTICAL 1

Theory:

DevOps is a culture + set of practices that brings **software development (Dev)** and **IT operations (Ops)** together so teams can **build, test, release, and run software faster and more reliably**.

Principles of DevOps

1. Collaboration & Shared Ownership

- Dev and Ops stop working in silos
- Everyone owns quality, performance, and reliability
- Failures are shared, fixes are collaborative

2. Automation Everywhere

- Automate repetitive tasks (builds, tests, deployments)
- Reduce human error
- Faster, consistent releases

3. Continuous Improvement

- Release small changes frequently
- Learn from failures (blameless postmortems)
- Measure → improve → repeat

4. Customer-Centric Thinking

- Deliver value quickly
- Short feedback loops from users
- Features are useless if they can't be deployed or maintained

5. Infrastructure as Code (IaC)

- Servers, networks, configs are defined in code
- Version-controlled, repeatable, and auditable

Key DevOps Practices

1. Continuous Integration (CI)

- Developers frequently merge code into a shared repo
- Automated builds and tests run on every change
- Catches bugs early

Tools: GitHub Actions, Jenkins, GitLab CI, CircleCI

2. Continuous Delivery / Continuous Deployment (CD)

- Code is always in a deployable state
- Deployment is automated
- In Continuous *Deployment*, changes go live automatically

Tools: Argo CD, Spinnaker, Azure DevOps, GitHub Actions

3. Infrastructure as Code (IaC)

- Define infrastructure using code instead of manual setup

Tools: Terraform, AWS CloudFormation, Ansible

4. Monitoring & Logging

- Track system health, performance, and errors
- Detect issues before users complain 🤖

Tools: Prometheus, Grafana, ELK Stack, Datadog

5. Security (DevSecOps)

- Security built into the pipeline, not added at the end
- Automated vulnerability scans, secrets management

Tools: Snyk, Trivy, Vault, SonarQube

6. Containerization & Orchestration

- Package apps with everything they need
- Run consistently across environments

Tools: Docker, Kubernetes

DevOps Engineer: Role & Responsibilities

Think of a DevOps Engineer as a **multiplier**—they make dev teams faster and systems more reliable.

Core Responsibilities

1. Build & Maintain CI/CD Pipelines

- Automate builds, tests, and deployments
- Ensure fast and reliable releases

2. Manage Cloud Infrastructure

- Provision and manage resources (AWS, Azure, GCP)
- Optimize cost, performance, and scalability

3. Automate Everything

- Scripts for deployments, backups, monitoring
- Reduce manual operational work

4. Ensure Reliability & Availability

- Implement monitoring and alerting
- Handle incident response and root-cause analysis

5. Improve Security

- Secure pipelines and infrastructure
- Manage secrets, access control, compliance

6. Collaborate Across Teams

- Work closely with developers, QA, and security
- Promote DevOps culture and best practices

Skills a DevOps Engineer Typically Needs

Technical Skills

- Linux & networking basics
- Git & version control
- CI/CD tools
- Cloud platforms (AWS/GCP/Azure)
- Containers (Docker, Kubernetes)
- Scripting (Bash, Python)

Soft Skills (Underrated but crucial)

- Communication
- Problem-solving under pressure
- Systems thinking
- Teaching and mentoring

Case Study Scenario (Practical Example)

Company Background

A mid-sized e-commerce company releases new features once every two months due to manual testing and deployment.

Challenges

- Frequent deployment failures
- Long downtime during releases
- Poor coordination between teams

DevOps Implementation

- GitHub used for version control
- Jenkins implemented for CI/CD
- Docker used for containerization
- Kubernetes used for orchestration
- Prometheus and Grafana for monitoring

Results

- Release frequency improved from bi-monthly to weekly
- Deployment failures reduced significantly
- Faster recovery from failures
- Improved customer satisfaction

Conclusion

DevOps is not just a set of tools but a cultural shift that emphasizes collaboration, automation, and continuous improvement. By adopting DevOps principles and practices, organizations can deliver software faster, more reliably, and with higher quality. The DevOps Engineer plays a critical role in enabling this transformation by designing efficient pipelines, automating infrastructure, and ensuring system reliability.